A HUMAN-IN-THE-LOOP MULTI-AGENT SWARM FOR SCIENTIFIC DISCOVERY IN THE PHYSICAL SCIENCES

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ABSTRACT

The automation of scientific discovery has seen rapid advancement, yet current end-to-end artificial intelligence (AI) systems remain largely confined to purely digital disciplines, such as machine learning and software engineering. These systems struggle to operate in the physical sciences, such as biophysics, materials science, and wet-lab biology, where experimental execution cannot be entirely simulated in a software sandbox. In this project, we hope introduce an AI-Collaborator, an open-source, human-in-the-loop system designed to augment the research pipeline for the physical and biological sciences. The AI-Collaborator employs a multi-agent swarm architecture to autonomously execute the cognitive heavy-lifting of the scientific method. The pipeline initiates with an evolutionary hypothesis generator governed by an automated LLM-as-a-judge tournament, followed by an exhaustive graph-traversal literature review. Crucially, methodology design is delegated to an asynchronous hive of specialized agents (theoreticians, experimentalists, and literature scouts) utilizing shared persistent memory to debate and refine rigorous, physically realizable protocols. Finally, the system automatically typesets its findings into a compiled master draft accompanied by automated peer-review comments. This project could be extended to other fields and may provide inspiration or assistance to scientific research. The open-source code: https://github.com/zhongya-lin/AI-Collaborator

Keywords AI · LLM · Natural Science

1 Introduction

The automation of scientific discovery has entered a paradigm-shifting era, driven by the rapid advancement of Large Language Models (LLMs) and autonomous multi-agent systems (Boiko et al. [2023], Wang et al. [2023]). Recent milestones, one of notably "the AI Scientist" (Lu et al. [2026]), have demonstrated that agentic workflows can autonomously generate hypotheses, write code, execute computational experiments, and compile research papers. However, a critical epistemic gap remains: these existing end-to-end systems are inherently confined to purely digital sandboxes (Yang et al. [2024]), such as machine learning, software engineering, or bioinformatics. In these domains, the entire scientific method, from ideation to execution and validation, can be simulated and verified within a closed computational loop.

This fully autonomous paradigm breaks down when applied to the physical sciences and biology, including biophysics, materials science, and wet-lab biology. In these fields, experimental execution cannot be entirely virtualized. For example, a hypothesis regarding the viscoelasticity of epithelial tissues or the strain gradients of advanced nanomaterials requires physical laboratory equipment, complex multiphysics simulations, and rigorous sample preparation protocols. Current AI systems are unable to perform well in these disciplines because they lack the ability to interact with the physical world, and their unconstrained outputs may propose physically impossible methodologies or hallucinate non-existent laboratory techniques.

*My current academic research focuses on mechanics and biophysics.

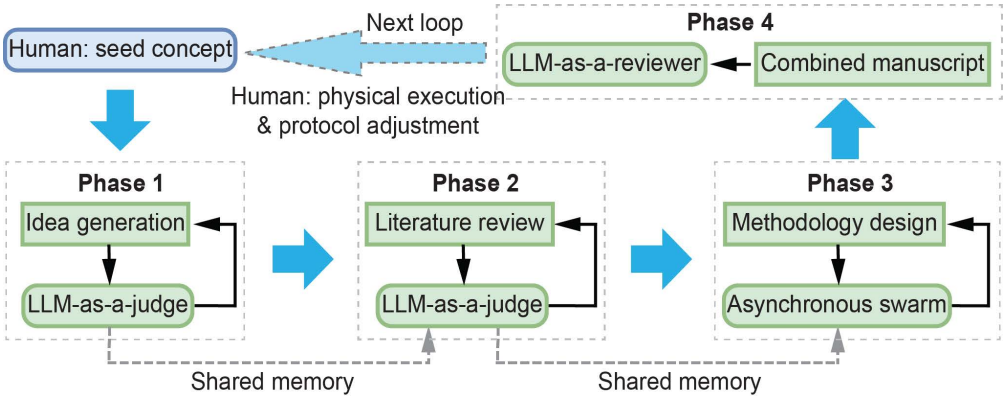

Figure 1: Schematic architecture of the AI-Collaborator framework. The pipeline initiates with a seed concept provided by the human researcher. Phase 1 generates candidate hypotheses evaluated via an evolutionary LLM-as-a-judge tournament. Phase 2 executes an exhaustive graph-traversal literature review to contextualize the scientific gap. Phase 3 utilizes an asynchronous multi-agent swarm to debate and synthesize the experimental and theoretical methodology. Dashed arrows indicate the flow of shared persistent memory, ensuring cross-phase consistency. In Phase 4, an agent compiles the artifacts into a master manuscript, which undergoes critical evaluation by an automated peer-review agent (LLM-as-a-reviewer). The final aligned draft and heuristic critiques are returned to the human researcher, bridging the digital-physical divide and informing the subsequent iteration of physical laboratory execution.

To address this limitation, we advocate for a fundamental shift in AI for science at current stage: transitioning from the illusion of a fully autonomous "AI Scientist" to the pragmatic reality of an AI-Collaborator. In this project, we introduce an open-source, human-in-the-loop multi-agent framework tailored specifically for the physical and biological sciences, but can be extend to other fields easily. Rather than attempting the impossible task of automating physical wet-lab execution and complicated physical simulations, the AI-Collaborator is designed to act as an intellectual catalyst. It augments the human principal investigator by rapidly expanding the boundaries of conventional ideation, breaking down literature silos, and accelerating the cognitive mapping of possible research pipelines.

The architecture achieves this through an evolutionary hypothesis generator, graph-traversal literature synthesis, and an asynchronous swarm of specialized agents (a theoretician, an experimentalist, and a literature scout) that iteratively debate and formulate methodological blueprints. We are explicitly honest about the system's boundaries: it does not produce verified empirical discoveries. Instead, its primary utility lies in generating novel, logically sound hypotheses, opening new scientific horizons, and architecting highly structured, physically plausible experimental and theoretical designs. The final output is a internally consistent master draft of a research proposal, accompanied by automated peer-review feedback. By constraining the AI to the realm of high-level ideation and methodology design, the AI-Collaborator bridges the critical gap between digital brainstorming and physical laboratory execution, serving as an elite, horizon-expanding co-pilot for human researchers.

2 System architecture

The AI-Collaborator operates as an asynchronous, multi-phase pipeline. To ensure the generated proposals remain physically plausible and scientifically rigorous, the architecture relies on strict data schemas (via Pydantic) and deterministic state-protector algorithms that intercept and sanitize complex scientific outputs before they trigger catastrophic parsing failures. The pipeline initiates with ideation and contextualization, executing in distinct phases before methodological design. The main architecture of this project is shown as Figure 1. In the following, we will take the topic of the application of strain gradient viscoelasticity (Lin and Wei [2020], Lin et al. [2026]) in biological research as an example, using Gemma4:26B locally.

2.1 Phase 1: Evolutionary idea generation

The scientific method begins with a testable hypothesis. However, LLMs prompted to generate "novel ideas" without structural constraints often produce superficial variations of highly cited papers, a phenomenon known as mode collapse. To counteract this, Phase 1 employs an evolutionary generation strategy governed by an LLM-as-a-judge tournament. The human researcher initiates the pipeline by providing a highly structured seed document containing a preliminary

title, keywords, and a brief abstract defining the core problem. Acting as a creative engine, the system stochastically generates a batch of distinct, expanded research proposals based on this seed.

Rather than arbitrarily selecting the first output, the system evaluates the batch using an automated LLM-as-a-judge. Each proposal is rigorously scored (e.g., out of 30 points) based on three criteria: Novelty (does it propose a new mechanism rather than merely observing a known correlation?), Feasibility (can it be executed using standard physical/biological laboratory equipment?), and Impact (does it address a fundamental gap suitable for high-tier publication?). The system outputs a fully ranked JSON array of the proposals alongside detailed justifications for their scores. Crucially, this establishes the first "Human-in-the-Loop" checkpoint, as the automated score is not treated as an absolute ground truth. The hypothesis that achieves the highest AI-generated score is not necessarily the optimal choice for physical execution. LLM evaluators frequently overvalue theoretical complexity while underestimating nuanced physical constraints, such as the availability of highly specific imaging modalities or the degradation rates of biological samples. Therefore, the AI-Collaborator relies on the human researcher to make the final heuristic judgment. As illustrated in Figure 2, the human researcher may review the tournament logs and explicitly reject the AI's top-ranked preference (Figure 2a) in favor of a runner-up (Figure 2b). By elevating a lower-ranked idea that presents a superior, physically realizable risk-reward profile tailored to their specific laboratory, the researcher can bridge the gap between digital ideation and physical reality.

2.2 Phase 2: Graph-traversal literature synthesis

A hypothesis is only valid if it accurately addresses a verified gap in the existing literature. Once the primary hypothesis is selected, it is passed to the Literature Scout, a specialized agent tasked with traversing the academic graph. Integrating directly with the Semantic Scholar API (Kinney et al. [2023]), the Scout formulates specific search queries to retrieve relevant peer-reviewed literature. However, a possible limitation of using LLMs to interface with academic APIs is the "keyword paradox": LLMs tend to generate overly specific, natural-language search queries (e.g., "mechanosignature decoding quantifying rate-dependent rheology") which cause strict API endpoints to return zero results.

To ensure resilience, the Literature Scout is equipped with a deterministic, self-healing search algorithm. The system autonomously truncates the query to broader foundational keywords, applies URL-encoding, and executes a recursive search with exponential backoff to bypass rate limits. Once the relevant papers are retrieved, the Scout synthesizes a comprehensive markdown document. This document does not merely summarize the papers, as shown in Figure 3. It actively contrasts the retrieved literature against the winning hypothesis, explicitly mapping the "Related Work" to identify the precise methodological or theoretical gap that the subsequent Swarm Phase must solve.

2.3 Phase 3: The methodology design

While generating a hypothesis and finding related literature can be achieved by sequential prompting, designing a rigorous, physically realizable experiment and simulation requires cross-disciplinary consensus. To achieve this, the AI-Collaborator deploys an asynchronous, multi-agent swarm—termed the "Hive"—consisting of specialized personas: a Theoretician, an Experimentalist, and a Literature Scout. These agents operate within a shared, persistent memory environment, engaging in iterative, multi-turn debates to synthesize the final methodology. For example, if the Experimentalist proposes an in vitro traction force microscopy (TFM) setup to measure epithelial buckling, the Theoretician must formulate the corresponding continuum mechanics model (e.g., active stress tensors coupled with viscoelasticity). If the Theoretician proposes a mathematical parameter that cannot be empirically measured by the Experimentalist's equipment, the agents critique each other and revise the protocol until cross-disciplinary consensus is reached.

A representative snapshot of this methodology output is provided in Figure 4. While the system identifies and aggregates relevant physical and computational methods into a research framework, it is important to critically evaluate the nature of this generated text. We observe that the generation occasionally exhibits methodological verbosity; rather than proposing a highly constrained, minimalist experimental design, the LLMs tend to over-generate, proposing an exhaustive array of techniques and variables. Nevertheless, this output firmly validates the capacity of multi-agent systems to synthesize complex, cross-disciplinary scientific protocols. Furthermore, this tendency to over-generate is not an inherent flaw of the architecture, but rather a parameter that can be optimized. In future iterations, this verbosity could be mitigated by employing more rigorously constrained prompt engineering and imposing stricter token-length penalties during the initial debate phase of agents .

2.4 Phase 4: Alignment and automated peer review

In human academic pipelines, a lead author or senior researcher must review individual contributions from collaborators to ensure cohesive logic and standardized terminology across the manuscript. The AI-Collaborator mirrors this process

(a)


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"Name": "Information-Theoretic Gradient Mechanics (ITGM)",
"Title": "Information-Theoretic Gradient Mechanics (ITGM): A New Paradigm for Spatiotemporal Self-Organization in Advanced Materials",
"Short Hypothesis": "The spatial gradient of the strain field ($\\nabla \\epsilon$) generates a 'Mechanic-Informational Potential' ($\\mu_{\\mathcal{I}}$) derived from
the gradient of the microstructural Fisher information, driving a directed flux of defects and molecules that governs the spatiotemporal self-organization of advanced
materials.",
"Related Work": "Existing frameworks like Morphogenetic Gradient Mechanics (MGM) and Topological Strain-Scale Mechanics (TSGM) focus on the physical and topological
consequences of strain gradients, such as flux-driven remodeling or node reconfiguration. However, they treat the material's microstructural state as a purely physical
or topological entity, neglecting the role of 'structural uncertainty' or 'information density.' ITGM distinguishes itself by introducing the Fisher information of the
microstructural state as a fundamental thermodynamic driver, proposing that the gradient of mechanical information ($\\nabla \\mathcal{I}$) creates a potent force for
the directed transport of mass and defects, providing a unified information-theoretic basis for cross-scale self-organization.",
"Abstract": "The ability to predict and control the spatiotemporal evolution of advanced materials\u2014ranging from nanocrystalline metals to biological tissues\u2014is
fundamental to the next generation of engineering and biotechnology. While current strain-gradient theories have successfully addressed size-dependent mechanics and
structural remodeling, they are primarily descriptive of the mechanical or topological consequences of deformation. This proposal introduces a transformative paradigm:
Information-Theoretic Gradient Mechanics (ITGM). We hypothesize that the spatial gradient of the strain field ($\\nabla \\epsilon$) creates a 'Mechanic-Informational
Potential' ($\\mu_{\\mathcal{I}}$) by modulating the Fisher information of the microstructural state. This gradient in 'structural uncertainty' acts as a thermodynamic
driver for the directed flux of microstructural constituents, such as dislocations in alloys or signaling proteins in the extracellular matrix (ECM). This mechanism,
which we term 'Information-Driven Mass Transport' (IDMT), provides a unified physical basis for the self-organizing properties of complex, heterogeneous materials. We
will validate this via two disparate systems: (i) mapping the correlation between dislocation-density-driven information gradients and solute-atom flux in
nanocrystalline Ni via 4D-STEM and Atom Probe Tomography, and (ii) observing the redistribution of proteins in a collagenous hydrogel subjected to microfluidic-generated
strain gradients. By integrating this theory into a stochastic phase-field framework, this work aims to establish a new mathematical foundation for 'information-encoded'
materials, enabling the design of autonomous, adaptive structures capable of programmed, gradient-responsive reconfiguration.",
"Experiments": "1. **Nanocrystalline Alloy Analysis (The 'Metal' Test):** Using high-resolution 4D-STEM, we will map the strain-induced distribution of grain
orientations and dislocation densities in a Ni-Cu nanocrystalline alloy under a controlled-gradient indentation. By coupling this with Atom Probe Tomography (APT), we
will quantify whether the gradient of the structural Fisher information ($\\nabla \\mathcal{I}$) correlates with the flux of Cu solute atoms toward or away from the
high-gradient zone. 2. **Bio-Hydrogel Remodeling (The 'Tissue' Test):** Using a microfluidic-based strain-gradient generator, we will subject a collagen-hyaluronic acid
hydrogel to a localized $\\nabla \\epsilon$ field. We will track the spatiotemporal redistribution of fluorescently labeled, size-tunable proteins using confocal
microscopy to determine if the 'information-driven' flux follows the predicted $\\mu_{\\mathcal{I}}$ potential, effectively testing the limits of mechanosensitive mass
transport.",
"Simulations": "We will develop a 'Stochastic-Gradient-Continuum' (SGC) framework. This framework will integrate a modified Phase-Field model where the free energy
functional $\\mathcal{F}$ is augmented by a Fisher Information term: $\\mathcal{F} = \\int [ \\psi(\\boldsymbol{\\epsilon}) + \\frac{1}{2}\\kappa |\\nabla \\boldsymbol
{\\epsilon}|^2 + \\lambda \\mathcal{I}(\\boldsymbol{\\epsilon}) ] d\\Omega$. The evolution of the microstructural order parameter will be governed by a modified
Cahn-Hilliard equation, where the chemical potential $\\mu = \\delta \\mathcal{F}/\\delta \\phi$ includes a term proportional to the gradient of the structural
information density, allowing for the simulation of information-driven pattern formation.",
"Limitations": "The primary challenge lies in the experimental quantification of 'structural information' ($\\mathcal{I}$), which requires extremely high-resolution,
statistically significant datasets (e.g., from 4D-STEM). Additionally, decoupling the information-driven flux from concurrent classical mechanical-driven fluxes (e.g.,
purely stress-driven diffusion) requires highly controlled, near-equilibrium loading protocols. Computationally, the inclusion of high-order Fisher information terms in
a continuum framework significantly increases the complexity of the partial differential equations and the required spatial resolution.",
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(b)


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"Name": "Spatiotemporal Gradient-Induced Viscoelasticity (SGIV)",
"Title": "Spatiotemporal Coupling of Strain Gradients and Viscoelastic Relaxation in Morphing Bio-Hybrid Architectures",
"Short Hypothesis": "Spatial strain gradients ($\nabla \\epsilon$) act as a driving force for the temporal evolution of the material's relaxation modulus, effectively
coupling the internal length scale of strain gradient theories with the temporal relaxation spectrum of viscoelasticity.",
"Abstract": "The ability to predict the long-term mechanical stability of advanced materials, from nanocrystalline alloys to biological tissues, requires a unified
spatiotemporal framework. While strain gradient theories have successfully captured spatial size effects (the 'where'), and viscoelasticity has captured temporal
relaxation (the 'when'), the fundamental coupling between spatial gradients and temporal evolution remains unexplored. We propose a new theoretical paradigm:
Spatiotemporal Gradient-Induced Viscoelasticity (SGIV). We hypothesize that a spatial gradient in strain ($\nabla \\epsilon$) induces a localized, time-dependent
evolution of the material's relaxation spectrum $E(t, \nabla \\epsilon)$ by driving the migration or reconfiguration of internal structural elements. This mechanism,
which we term 'Gradient-Driven Viscoelastic Aging' (GDVA), provides a physical basis for the observed spatiotemporal heterogeneity in tissue remodeling and the anomalous
relaxation observed in gradient-strained polymers. To validate this, we will employ a dual-approach: (i) using microfluidic-based strain-gradient microrheology to
observe the temporal evolution of the complex modulus in a gradient-controlled hydrogel, and (ii) developing a new constitutive framework that integrates a
gradient-dependent viscosity kernel into a non-local viscoelastic-plasticity model. This work promises to provide the mathematical foundation for designing 'adaptive'
materials that can program their mechanical properties in response to localized mechanical stimuli.",
"Experiments": "1. Microfluidic Gradient-Strain Device: Design a microfluidic chip capable of generating a controlled, steady-state strain gradient across a viscoelastic
hydrogel (e.g., PEG-DA or collagen). 2. Spatiotemporal Microrheology: Embed fluorescently labeled probe particles within the gradient. Use high-speed confocal microscopy
and Particle Tracking Microrheology (PTM) to measure the local Mean Squared Displacement (MSD) as a function of both position (x) and time (t). 3. Fluorescence Recovery
After Photobleaching (FRAP): Use FRAP to track the diffusion and redistribution of crosslinking molecules within the high-gradient and low-gradient regions to correlate
structural rearrangement with the measured $\nabla E(t)$.",
"Simulations": "Development of a New Constitutive Law: A Finite Element Method (FEM) implementation of a non-local, time-dependent viscoelastic model where the viscosity
$\\eta$ is a function of the strain gradient: $\\eta = \\eta_0 [1 + \\alpha (\nabla \\epsilon)^2]$. The simulation will model a 'morphing' geometry, predicting how a
spatial gradient in loading leads to a spatially heterogeneous 'stiffening' or 'softening' of the material over long temporal scales.",
"Limitations": "1. Experimental Resolution: The ability to resolve very high-frequency strain gradients in highly dissipative media is limited by the particle-tracking
precision. 2. Computational Cost: Implementing non-local, time-dependent kernels in 3D FEM is computationally expensive and may require significant simplification (e.g.,
using a gradient-expansion approach). 3. Material Complexity: Real biological tissues involve active biological processes (cell-driven) which may mask the purely
physical 'gradient-driven' aging effect.",
"Related Work": "Classical strain gradient plasticity (SGP) accounts for size effects via an internal length scale $\\ell$ but treats time-dependent relaxation as a
separate, local phenomenon. Viscoelasticity models describe temporal decay but ignore the influence of $\nabla \\epsilon$ on the rate of relaxation. This proposal
introduces a 'Gradient-Driven Viscoelastic Aging' (GDVA) mechanism, where the spatial gradient of deformation triggers a non-local redistribution of dissipative elements
(e.g., polymer entanglements or protein crosslinks) over time.",
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Figure 2: The generated ideas. (a) The proposal that achieved the highest automated score from the LLM-as-a-judge out of 20 generated candidates. (b) The runner-up proposal (ranked 4th overall) that was ultimately selected by the human researcher.

Related Work

The conceptual foundations of Information-Theoretic Gradient Mechanics (ITGM) are situated at the intersection of information theory, statistical mechanics, and non-equilibrium thermodynamics. While the drive to treat information as a physical constituent of reality is well-documented—with information-theoretic interpretations of quantum mechanics suggesting that the fundamental structure of physical laws may be inherently epistemic [3957...], [77de...]—the application of these principles to the mechanical evolution of matter remains largely unexplored. The development of ITGM seeks to bridge the gap between the mathematical quantification of structural complexity and the physical manifestation of material self-organization.

2.1. Information Theory as a Metric for Stochastic Dynamics Current literature extensively utilizes information theory to quantify the generalization capabilities and complexity of stochastic processes. In the context of machine learning, information-theoretic frameworks have been established to derive generalization bounds using Integral Probability Metrics (IPM) and f-divergences, including Wasserstein and KL-divergence bounds [28dea72fc1ec3933c858a473afc2d2cfac]. Furthermore, research has demonstrated how the local statistics of stochastic gradients—specifically variance and smoothness—influence the convergence and generalization error in Stochastic Gradient Descent (SGD) [f6d9da39f7f2431b99b5a373ef1efeabd188e955].

However, a critical limitation exists: these frameworks are primarily focused on *error estimation* and *learning complexity* rather than the *physical mechanics* of material assembly. It has been demonstrated that many existing frameworks, including those based on mutual information and PAC-Bayes, fail to establish minimax rates for gradient descent in stochastic convex optimization [9a2ddc9c5791057d033fe94ab70fcf7c1208658f]. While neural networks can leverage higher-order information to learn complex functions [b92985bc6723edaf7c450b970751602552a67a0d], these approaches treat information as a *metric* to bound uncertainty, rather than a *constitutive driver* capable of exerting physical force.

2.2. Statistical Mechanics and Information-Driven Transport The bridge between information and physical transport is well-established in molecular communication (MC) systems. The Langevin equation serves as a unified modeling tool for molecule propagation, encompassing random walks, drifted random walks, and active transport [df47bd672032bd9551c1d39e6cd1b628bda97fed]. This aligns with the broader pursuit of integrating predictive statistical mechanics to understand complex, non-equilibrium systems [831-4120d514ed004ef20d59ed76bdac19e82b929]. In biological systems, such as bacterial colonies, spatiotemporal self-organization emerges from the coupling of motility and population density [4be6...].

While these works describe how information is transmitted via molecular exchange or how entropy gradients can drive thermal effects in extreme gravitational environments [c0cb...], they do not yet treat the *spatial gradient of the mechanical strain field* ($\nabla\epsilon$) as a primary source of a Mechanic-Informative Potential ($\mu_{\mathcal{I}}$). Current models of transport are largely driven by concentration or density gradients; ITGM proposes that the gradient of the microstructural Fisher information creates a potent, independent force for Information-Driven Mass Transport (IDMT).

2.3. Strain-Gradient Mechanics and Spatiotemporal Self-Organization In materials science, the study of spatiotemporal self-organization has traditionally focused on the physical and topological consequences of deformation. Existing frameworks, such as Morphogenetic Gradient Mechanics (MGM) and Topological Strain-Scale Mechanics (TSGM), have successfully addressed size-dependent effects and node reconfiguration in response to strain. More recently, "Thermodynamic Consistent Neural Networks" (TCNN) have been introduced to model interfacial mechanics, ensuring that data-driven models obey physical laws such as energy dissipation [4732cbe3699535721ce9e25bd65f9b552e32668f].

Despite these advancements, current strain-gradient theories are primarily *descriptive* or *reactive*—they model the mechanical consequences of deformation or enforce existing thermodynamic laws, treating the microstructural state as a purely physical or topological entity. They neglect the role of "structural uncertainty" or "information density" as a fundamental driver. While studies on symmetry breaking suggest that gradients are fundamental to new structural regimes [6f67...], we lack a constitutive law that links the mechanical strain field ($\nabla\epsilon$) to an informational potential ($\mu_{\mathcal{I}}$) that can actively steer the flux of mass and defects.

Figure 3: Automated literature synthesis generated during Phase 2. Based on the highest-scoring hypothesis from the Phase 1 tournament, the Literature Scout agent retrieves relevant peer-reviewed publications and synthesizes them into a cohesive review.


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"Experimental_Protocol":
    {"Objective": "To validate the existence of Information-Driven Mass Transport (IDMT) by demonstrating that mass flux is driven by gradients
    in microstructural topological Fisher Information ($\nabla \\mathcal{I}$) rather than traditional chemical or thermal gradients.",
    "System_I_Nanocrystalline_Ni": {"Sample_Preparation": "Utilize a Cryo-FIB (Focused Ion Beam) lift-out protocol on single-needle specimens to
    prevent vacancy-driven solute flux and preserve the transient microstructural state from ion-beam-induced diffusion.",
    "Characterization_Suite": {"Strain_and_Topology_Mapping": "High-resolution 4D-STEM using Direct Electron Detectors (DED) in
    electron-counting mode. Implementation of a low-dose imaging regime is mandatory to prevent beam-induced dislocation motion. The goal is to
    resolve the topological configuration density ($\theta$) of dislocation loops.", "Chemical_Gradient_Mapping": "Atom Probe Tomography (APT)
    to map solute-atom distribution with atomic precision.", "Correlative_Protocol": "Implementation of a 'Correlative APT-TEM' protocol on the
    same single-needle specimen to ensure spatial registration of the strain gradient ($\nabla \\mathcal{\\epsilon}$) and the chemical gradient
    ($\nabla c$)."}, "Data_Validation_Metrics": {"Information_Partitioning": "Explicitly partition Fisher Information into Aleatory (intrinsic
    microstructural fluctuations) and Epistemic (instrumental/measurement) components using a known-standard single-crystal lattice calibration.
    ", "Noise_Invariance_Test": "Subject reconstructed 4D-STEM images to controlled Gaussian noise degradation; if the observed flux $\\mathcal
    {J}$ vanishes, the effect is flagged as an epistemic artifact."}}, "System_II_Collagenous_Hydrogel": {"Sample_Preparation":
    "Microfluidic-generated controlled strain gradients applied to 3D collagenous matrices.", "Characterization_Suite":
    {"Volumetric_Strain_Mapping": "Transition from 2D Traction Force Microscopy (TFM) to 3D Digital Volume Correlation (DVC) enabled by
    Light-Sheet Fluorescence Microscopy (LSFM) to resolve the internal 3D strain tensor ($\\epsilon_{ij}$) and its gradient.",
    "Protein_Redistribution_Tracking": "Fluorescently labeled signaling proteins to track mass flux $\\mathcal{J}$ in response to $\nabla
    \\epsilon$."}}},
"Simulation_Framework": {
    "Mathematical_Foundation": {"Energy_Functional": "A unified thermodynamic framework based on the variational derivative of a total free
    energy functional $\\mathcal{F}_{total} = \\mathcal{F}_{chem} + \\mathcal{F}_{elastic} + \\mathcal{F}_{info} + \\mathcal{F}_{reg}$, where
    $\\mathcal{F}_{info}$ represents the energy density contribution of the Fisher Information.", "Topological_Chemical_Coupling_TCC": "To
    ensure the Mechanically-Informational Potential ($\\mu_{\\mathcal{I}}$) acts on mass flux, the parameters of the microstructural
    distribution $\theta$ (e.rin., dislocation loop radii) must be explicitly defined as functions of the local concentration $c$."},
    "Numerical_Implementation": {"Kernel_Smoothed_Fisher_Information_KSFI": "To prevent 4th-order PDE singularities and the Gibbs phenomenon,
    $\\mathcal{I}$ will be calculated via a Gaussian-kernel convolution: $\\mathcal{I}_\\sigma = \\int p(\theta) \\mathcal{G}_\\sigma(\theta -
    \theta') d\theta'$.", "Scale_Consistency": "The smoothing kernel width $\\sigma$ must be physically coupled to the intrinsic microstructural
    correlation length $\\xi$ (e.g., the dislocation core radius) to avoid arbitrary hyperparameter dependence.",
    "Adaptive_Mesh_Refinement_IT_AMR": "Implementation of an 'Information-Triggered' AMR strategy, where mesh density is dynamically increased
    in regions of high $\nabla \\mathcal{I}$ to resolve high-frequency topological gradients without prohibitive global computational costs.",
    "Regularization": "Inclusion of a Cahn-Hilliard-type gradient-regularization term ($\\kappa|\nabla \\phi|^2$) to prevent numerical
    shattering and unphysical microstructural oscillations."}
```


Figure 4: Automated methodology blueprint generated during Phase 3. The Swarm synthesizes the selected hypothesis and literature context to output a comprehensive experimental and theoretical protocol.

Table 1: LLM used in this project

Name	Description	Cost per idea
Deepseek v3.2	With more than 600 billion total parameters in its enhanced Mixture of Experts design and 37 billion activated per token (Liu et al. [2025])	less than $0.1
Gemma4	With $\sim$ 4 billion active parameters (Team et al. [2024])	free

through an Editor-in-Chief alignment engine.The Editor-in-Chief ingests the three disparate artifacts—the selected Hypothesis, the Literature Review, and the Methodology, and performs a comprehensive consistency check. It detects and resolves logical contradictions (e.g., ensuring the literature gap perfectly aligns with the proposed experimental boundary conditions) and standardizes mathematical variables across the text.

Crucially, the final output of the AI-Collaborator is not presented as a "camera-ready" scientific discovery. Instead, the system outputs the aligned master draft alongside automated peer-review comments (Rao et al. [2025]). The system initiates a final evaluation round where an independent LLM agent, prompted as a critical reviewer, aggressively critiques the generated proposal. It identifies potential failure modes in the experimental design, challenges the assumptions of the theoretical model, and highlights unaddressed limitations.By delivering a highly structured proposal coupled with a pre-emptive peer-review critique, the AI-Collaborator provides the human researcher with an actionable, heavily scrutinized blueprint. This output directly accelerates the human-led phase of research, allowing the laboratory team to bypass weeks of preliminary brainstorming and immediately begin refining the physical execution of the experiments.

For now, we are only testing two LLMs: Deepseek Reasoner (Deepseek v3.2) and Gemma 4 (26b and e4b), as shown in Table 1.

3 Human-in-the-Loop execution: Bridging the digital-physical divide

A fundamental limitation of fully autonomous scientific agents is the "epistemic boundary" of the digital sandbox. AI cannot pipette reagents, calibrate an atomic force microscope, or synthesize a nanomaterial. The AI-Collaborator deliberately halts at this boundary, positioning the human researcher and the laboratory team as the essential physical actuators of the scientific method.

The transition from digital ideation to physical execution is formalized through the system's generation of actionable blueprints. Rather than outputting abstract conceptual summaries, the AI-Collaborator's Swarm methodology translates the theoretical consensus into precise experimental parameters and simulation frameworks. For computational physics, this includes defining specific boundary conditions, mesh granularities, and constitutive equations (e.g., finite element method, phase-field models) required to initialize the simulation. For physical wet-lab biology or materials science, the output serves as a high-fidelity foundation for Standard Operating Procedures, specifying necessary imaging modalities, cell lines, substrate stiffness gradients, or pharmacological inhibitors.

Crucially, the "Human-in-the-Loop" architecture is not a passive handoff; it is an active, iterative integration. When the human researcher receives the aligned master draft, they must concurrently evaluate the AI-generated reviewer critique. This automated peer review serves as a heuristic filter, explicitly warning the human team of potential physical failure modes, such as a proposed traction force microscopy setup lacking the temporal resolution required to capture the proposed viscoelastic relaxation times. The human researcher uses this critique to preemptively adjust the physical protocol before expending valuable laboratory resources.

Furthermore, this architecture enables a powerful asynchronous feedback loop. If the human laboratory team attempts the physical execution and encounters an unforeseen physical constraint, such as a specific polymer degrading under the proposed experimental temperature, the researcher does not need to abandon the hypothesis. Instead, the researcher can re-initialize the Methodology, injecting the newly discovered physical constraint into the system's persistent memory. The Theoretician and Experimentalist agents will then re-debate the methodology, generating an updated, constraint-aware experimental protocol. By utilizing the AI as a rapid, adaptive cognitive engine, the human team remains firmly in control of empirical validation, accelerating the pace of discovery while maintaining absolute physical rigor.

4 Limitations and future directions

While the AI-Collaborator tries to bridge the ideation-execution gap, deploying asynchronous multi-agent systems in the physical sciences presents distinct challenges that require ongoing refinement:

Hallucinations and epistemic risk The primary limitation of any LLM-driven architecture is semantic hallucination. While our prompts and algorithms eradicate syntactic crashes, the agent may still propose methodologies that are mathematically sound but physically impossible, such as assuming infinite boundary stiffness in a hydrogel or citing a hallucinated fluorophore. The Reviewer agent mitigates this by flagging theoretical inconsistencies, but the human researcher remains the ultimate epistemic failsafe. The system relies entirely on the human team to recognize and discard physically unrealizable parameters before laboratory execution.

Computational overhead and API bottlenecks Simulating multi-agent debate and executing exhaustive graph-traversal literature reviews is computationally intensive. Compared to single-shot prompting, the asynchronous Swarm consumes significantly more tokens and requires robust API error-handling (e.g., exponential backoff) to navigate strict academic database rate limits. As the complexity of the research topic scales, such as modeling cross-scale spatiotemporal dynamics from nanocrystalline structures to biological tissues, the token context windows required to maintain coherent memory across the Swarm become a limiting factor.

Architectural modularity and fine-tuned agents A distinct advantage of the AI-Collaborator's design is its highly modular architecture. Because each phase of the pipeline, i.e., ideation, literature traversal, methodology debate, and alignment, is relatively decoupled, the framework is not rigidly dependent on the current generation of general-purpose LLMs. As the field of open-source AI advances, the generalized agents currently driving the system can be systematically swapped with domain-specific, fine-tuned models (often referred to as specialized "skills"). For example, the general "Theoretician" agent could be seamlessly replaced by a specialized model explicitly fine-tuned on multiphysics simulation code, or the "Literature Scout" could be upgraded with an agent trained via reinforcement learning specifically for academic graph traversal. This modularity ensures that the AI-Collaborator framework remains a flexible, scalable infrastructure that can evolve in tandem with state-of-the-art advances in specialized AI models.

5 Conclusion

The pursuit of fully automated scientific discovery has historically been confined to the digital sandbox, restricted by the inability of algorithms to physically actuate experiments in the real world. In this project, we presented the AI-Collaborator: an open-source, human-in-the-loop multi-agent framework that explicitly acknowledges and leverages this physical boundary. By rejecting the paradigm of the autonomous "AI Scientist" in favor of an elite cognitive co-pilot, we demonstrated how an integrated system—featuring an evolutionary hypothesis tournament, self-healing literature traversal, an asynchronous Swarm, and an Editor-in-Chief alignment engine—can accelerate the most demanding phases of the scientific method. The AI-Collaborator assumes the cognitive heavy-lifting of cross-disciplinary ideation and methodological blueprinting, outputting rigorously aligned proposals alongside pre-emptive peer review. In doing so, it firmly establishes a highly synergistic future for the physical sciences: one where artificial intelligence acts as a relentless intellectual catalyst, while human researchers remain the ultimate masters of physical execution and empirical truth.

Acknowledgments and AI contribution statement

The author acknowledges the use of Gemini 3.1 Pro (a large language model built by Google (Team [2023])) as a computational assistant in the development of this AI-Collaborator project and the drafting of this manuscript. To ensure full academic transparency, the division of labor is explicitly defined as follows:

- Human (Author): Conceptualized the core architecture of the AI-Collaborator, formulated the paradigm shift from "AI Scientist" to "AI-Collaborator," developed the foundational Python scripts, and directed the intellectual narrative and scope of the manuscript.
- Artificial Intelligence (Gemini 3.1 Pro): Acted as a coding assistant to optimize the Python codebase (including the development of the deterministic JSON-LaTeX parsing algorithms), polished the academic prose, and assisted in structuring the narrative flow of the manuscript under the direct supervision and iterative prompting of the author.

The human author takes full responsibility for the concepts, code logic, and conclusions presented in this work.

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